CHAPTER 2:

Big Bang to Now

> Beginnings imply and require endings.
>
> Ann Leckie, *Ancillary Justice*

I love stories about time travel. It's easy to quibble about the physics of time machines or to balk at the various paradoxes that come up. But there's something appealing about the idea that we might somehow find a trick that will open up the past and future to our knowledge and intervention, to allow us to step off this runaway train of "now" barreling inexorably toward some unknown fate. Linear time just seems so restrictive, even wasteful—why should all that time, all those possibilities, be lost to us forever, just because the clock has ticked forward a few degrees? We may have grown accustomed to strict chronological oppression, but that doesn't mean we have to like it.

Fortunately, cosmology can help. Not in any practical sense, of course—we're still talking about a relatively esoteric branch of physics that will in no way enable you to get back the umbrella you left on the train yesterday. But rather, in the sense that your life remains the same but absolutely everything else about existence is forever changed.

To a cosmologist, the past is not some unreachable lost realm. It's an actual place, an observable region of the cosmos, and it's where we spend most of our workday. We can, while sitting quietly at our desks, watch the progress of astronomical events that happened millions or even billions of years ago. And the trick isn't special to cosmology, but inherent to the structure of the universe in which we live.

It all comes down to the fact that light takes time to travel. Light speed is fast—about 300 million meters per second—but it's not instantaneous. In everyday terms, when you switch on a flashlight, the light coming out of it covers about one foot per nanosecond, and the reflection of that light off whatever you're illuminating takes just as long to get back to you. In fact, when you look at anything, the image you see, which is just the light coming off it that reaches your eye, is a little bit stale by the time it gets to